

Problem M. Number of Islands Easy

OS Linux

You are given a 2D matrix consisting of the following:

1: land

0: water

Considering N-8 neighbourhood, your task is to find the number of islands in the given landscape.

Assume that the the 2D matrix is surrounded by water beyond the boundaries.

Definitions:

N-8 neighbourhood: 2 cells are considered neighbours, if they share an edge or a corner.

Island: A piece of connected land surrounded by water.

Input Format

First line of input contains T - number of test cases. The first line of each test case contains R, C - the size of the matrix. Its followed by a matrix of size RxC, consisting only of 0s and 1s.

Constraints

$1 \leq T \leq 100$

$1 \leq R, C \leq 500$

Output Format

For each test case, print the total number of islands, separated by new line.

Input	Output
3 4 8 01001101 10001010 00100100 01101101 6 1 0 1 0 0 1 1 9 5 00000 10001 00110 01100 10000 00000 00101 00011 00110	4 2 3

Explanation o

Self Explanatory